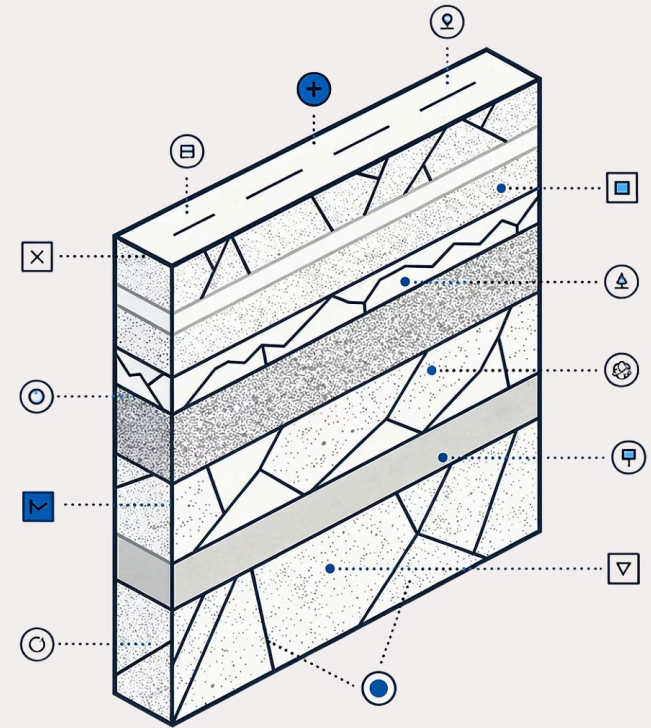


Paper 1 Writing Guide

*Bimodal Joint Population Characterisation at Najd-affected Road Cuts,
Makkah Region*



What This Paper Is – and Isn't

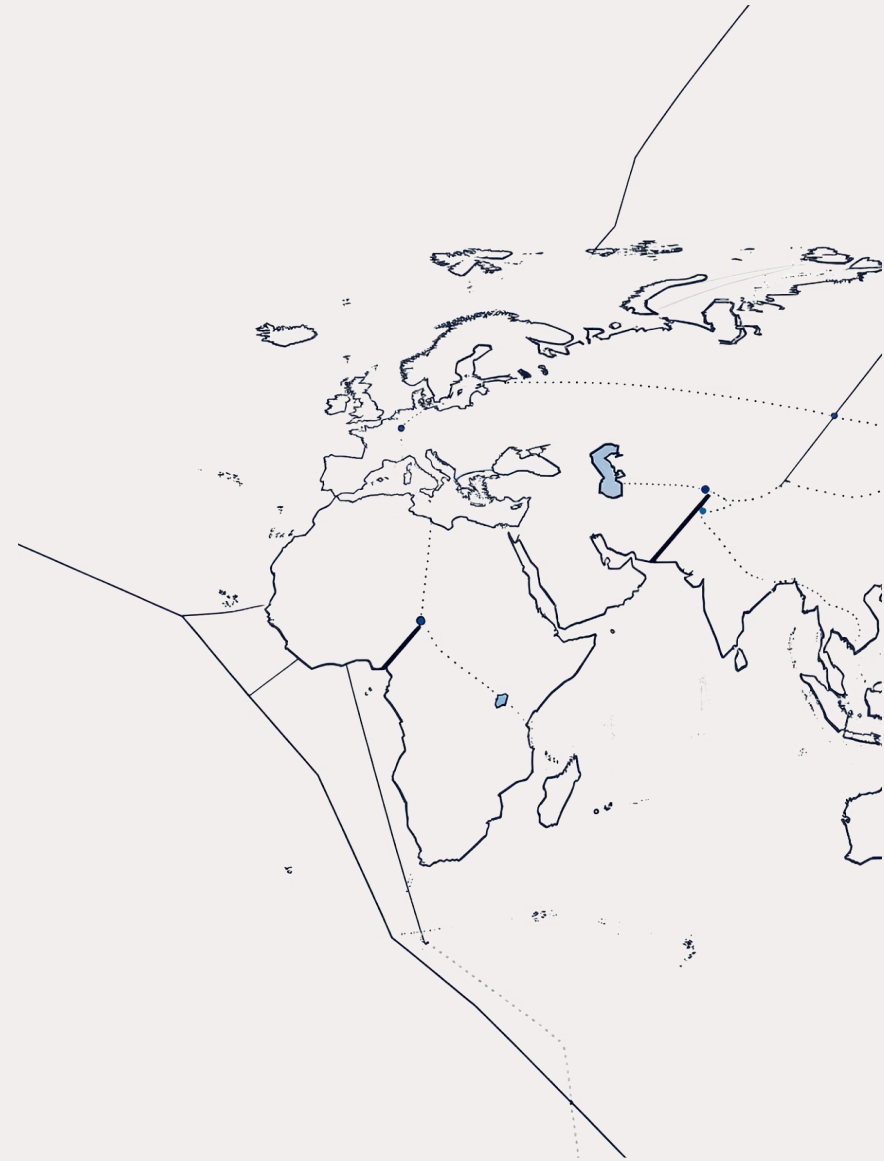
In Scope

- Statistical proof of two distinct joint families co-existing at all five sites
- Separation of Najd (NW–SE) and Red Sea (NE–SW) families by orientation and concentration
- Site-by-site mean direction, $\bar{R}$, and κ per family
- Geological interpretation of Site 2 overprinting and Site 5 three-way mixture

Out of Scope

- Najd Intensity Index → **Paper 2**
- Slope stability → **Paper 3**
- Risk mapping → **Papers 4–5**

 This paper establishes the quantitative foundation that all subsequent papers depend on.



Data at a Glance

330

Joint Measurements

Across 5 granodiorite road-cut outcrops,
Makkah Region

2

Joint Families

Najd (NW–SE, dominant) and Red Sea
(NE–SW, subordinate)

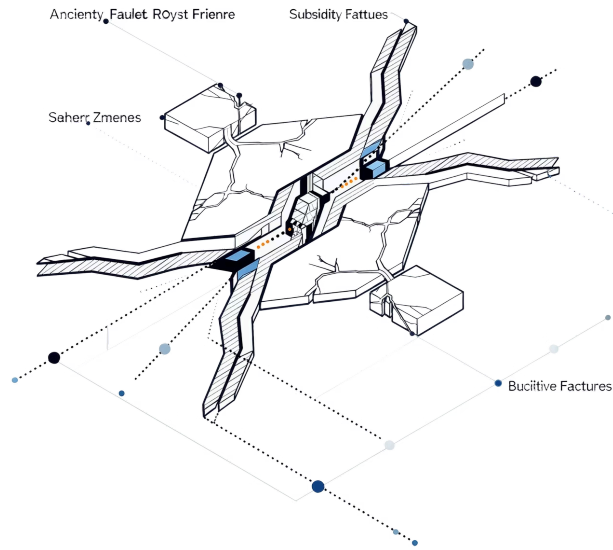
4/5

Sites Confirmed

Rayleigh $p < 0.05$ — non-uniformity
confirmed

✓ Lithological uniformity (all granodiorite) isolates structural from lithological variation — a key methodological strength.

Geological Setting: Two Tectonic Episodes



Najd Fault System (~600 Ma)

Proterozoic sinistral transpression; NW-SE shear corridor imprints the persistent NW-trending joint set.

Red Sea Rifting (~30 Ma)

Cenozoic NE-SW extension creates the subordinate but locally dominant NE-trending set.

Two episodes **~570 Ma apart** → bimodal fabric is geologically expected, not coincidental.

Statistical Methods



Rayleigh Test

Tests non-uniformity of azimuths; $Z = n \cdot \bar{R}^2$; $p < 0.05$ confirms a preferred direction.



V-test

One-tailed variant targeting a specific bearing (~135° Najd, ~050° Red Sea); useful where Rayleigh is borderline.



Von Mises Fitting

μ = mean direction, κ = concentration; **10 fits** (5 sites $\times$ 2 families).



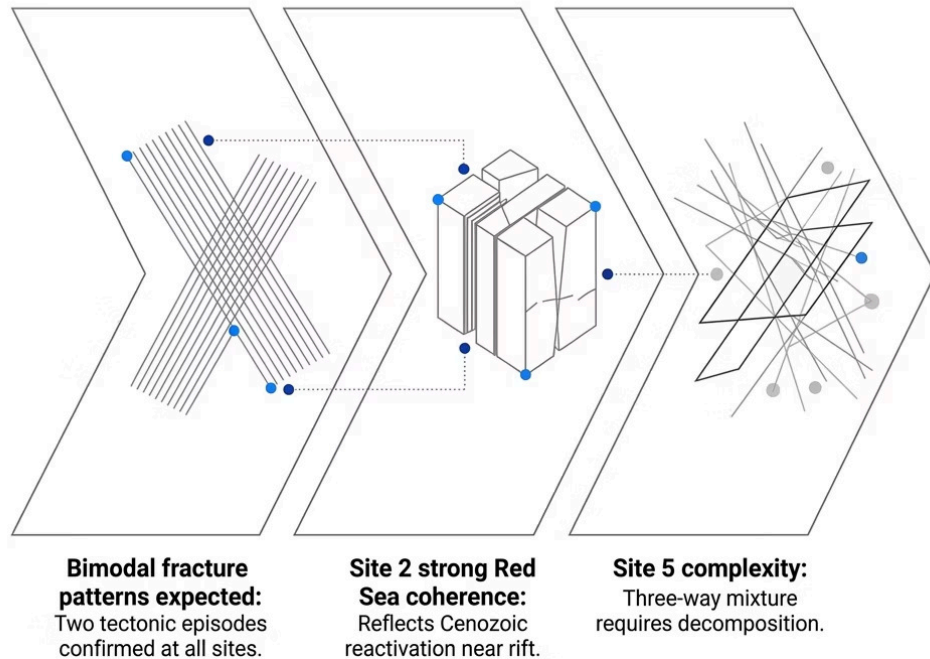
$\bar{R}$ Mean Resultant Length

Core diagnostic; Najd yields $\bar{R} = 0.98\text{--}1.00$ across sites — the key finding.

Key Results by Site

Site	Najd Family	Red Sea Family	Interpretation
1, 3, 4	$\bar{R} \approx 0.98\text{--}1.00$, $\mu \approx 135^\circ$	Subordinate; Rayleigh $p < 0.05$	Najd dominant — canonical bimodal pattern
2	Present but weaker	$\bar{R}$ exceeds Najd; $p < 0.05$	Cenozoic overprinting evident
5	39% of joints	18% of joints	Pooled Rayleigh $p = 0.228$; 42% unclassified — structural transition zone

Discussion: Three Interpretive Threads



Thread 1: Bimodal Expectation

Two independent tectonic episodes should leave two joint sets — the data confirms this at all five sites.

Thread 2: Site 2 Overprint

Stronger Red Sea coherence reflects Cenozoic reactivation or proximity to a rift-related structure.

Thread 3: Site 5 Complexity

Three-way mixture interpreted as a shear-zone boundary; family decomposition required before NII input.

Figures Checklist

1

Figure 1 — Location Map

Five outcrops with Najd corridor and Red Sea rift axis inset.

2

Figure 2 (2a–2e) — Rose Diagrams

Per site in 10° bins, colour-coded by family. Site 5 shows three-class decomposition — **the paper's core visual**.

3

Table 2 — Summary Statistics

n , $\bar{R}$, μ , κ , and Rayleigh p for each family at each site — must be internally consistent with rose diagrams.

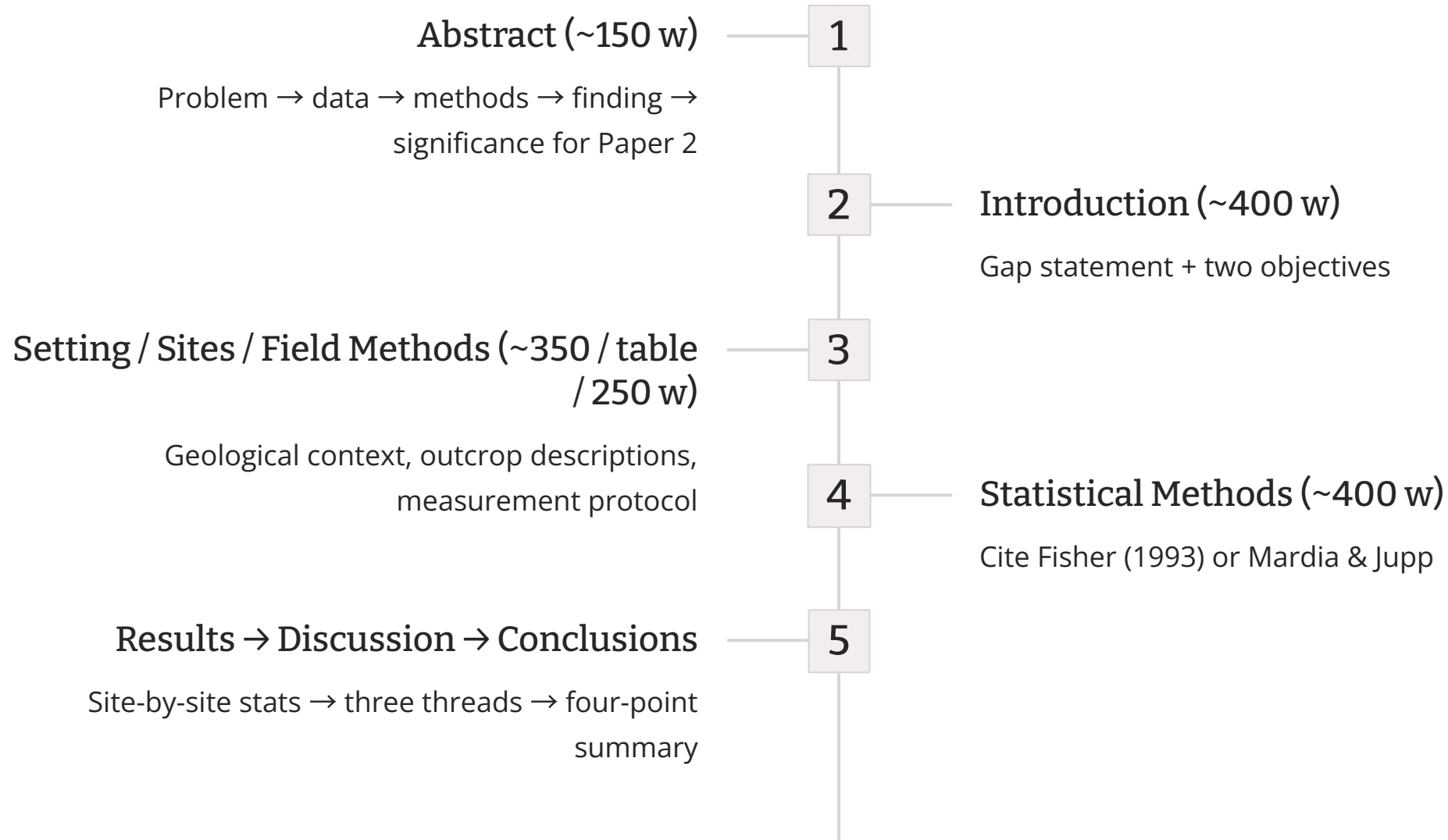
4

Figure 3 (Optional) — Stereonet

All 330 poles coloured by family — include if time allows.

⚠ Table 2 and Figure 2 must be finalised before drafting the Results section — internal consistency is non-negotiable.

Paper Structure & Word Targets



Conclusions: Four Points to Land

Non-Uniformity Confirmed

Non-uniform distribution confirmed at **4 of 5 sites** ($p < 0.05$).

Bimodal Fabric Universal

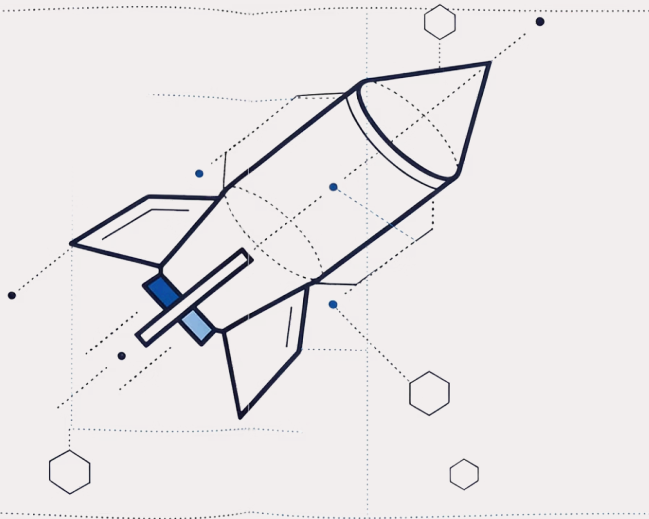
Bimodal fabric present at **all sites** — Najd NW–SE dominant, Red Sea NE–SW subordinate.

Site-Specific Anomalies

Site 2 Red Sea overprint; Site 5 structural complexity requiring family decomposition.

Foundation for Paper 2

Confirmed families form the **quantitative foundation for the Najd Intensity Index** in Paper 2.



Target: *Journal of Structural Geology*

Submission Timeline

3–5 months from now

Data collection complete —
ready to draft.

Priority Next Step

Finalise rose diagrams and Table 2
before writing the Results section.

- Internal consistency between figures and statistics is the critical gate before drafting begins.